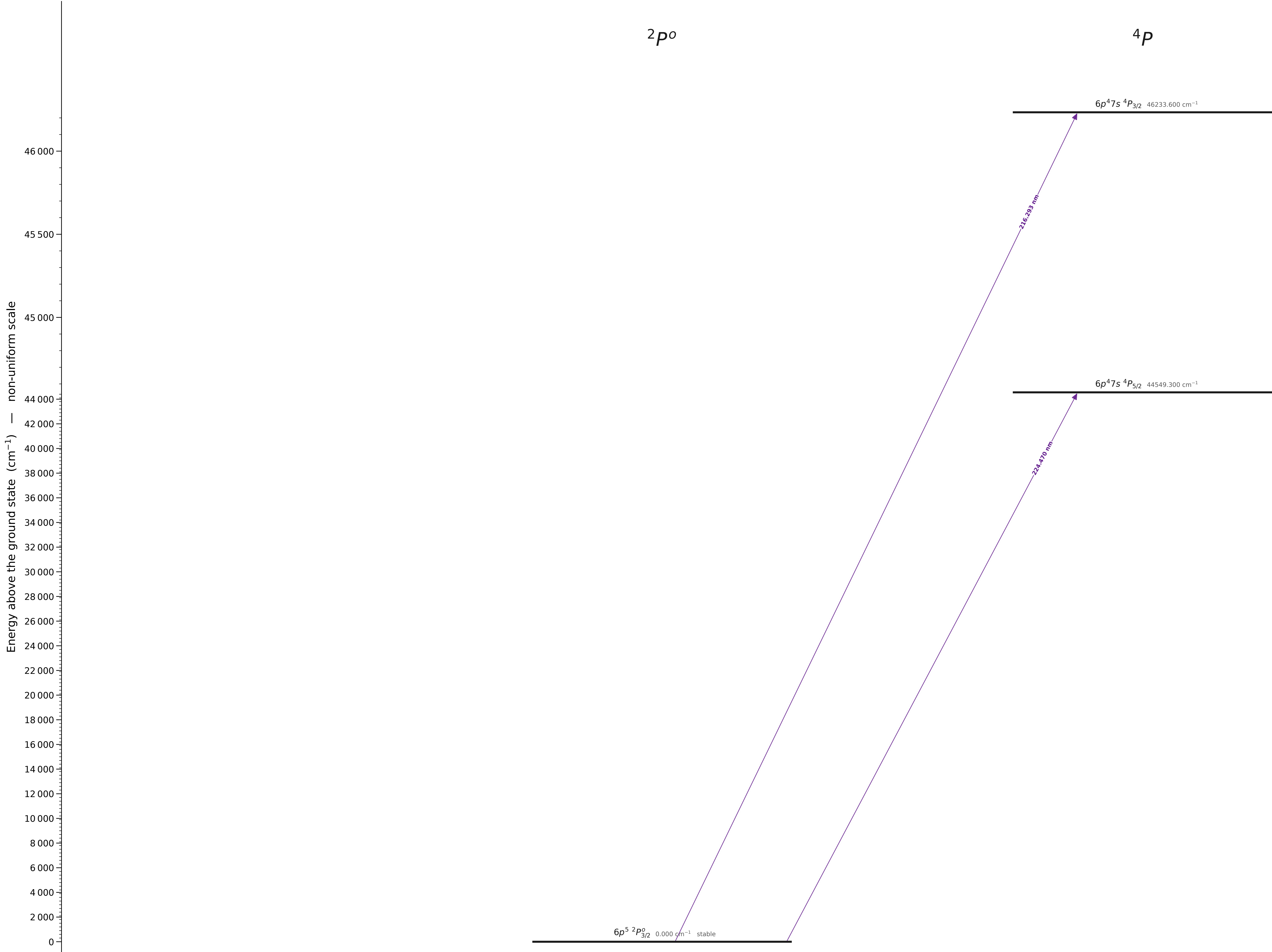
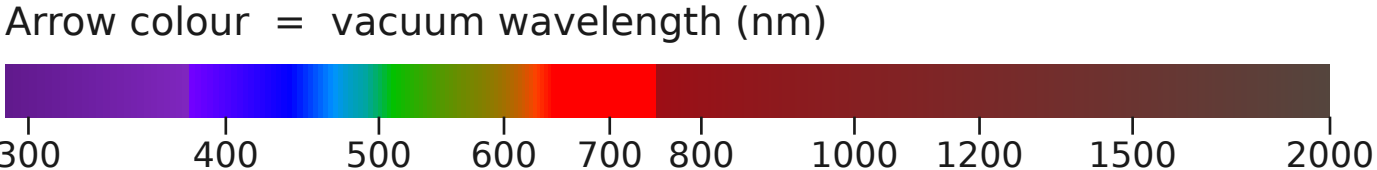


Astatine

At (At I). Strongest classified transitions below 2 μm (NIST): 3 levels, 2 lines.

At⁺ ionisation limit 75150.80 cm^{−1} = 9.31751 eV (133.066 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J \| e r \| J' \rangle |$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J \ e r \ J' \rangle $ (ea ₀)	A (s ^{−1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$6p^5\ ^2p^o_{3/2} - 6p^47s\ ^4p_{3/2}$	216.2929	216.2250	1386.0485	-	-	-	-	first step of the ISOLDE-RILIS / TRIUMF-TRILIS three-colo...
$6p^5\ ^2p^o_{3/2} - 6p^47s\ ^4p_{5/2}$	224.4704	224.4008	1335.5544	-	-	-	-	alternative first step for resonance ionization

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$6p^5\ ^2p^o_{3/2} - 6p^47s\ ^4p_{3/2}$	216.293	197-205	-7190(750)
$6p^5\ ^2p^o_{3/2} - 6p^47s\ ^4p_{3/2}$	216.293	197m-205	-2975(750)
$6p^5\ ^2p^o_{3/2} - 6p^47s\ ^4p_{3/2}$	216.293	198-205	-8731(750)
$6p^5\ ^2p^o_{3/2} - 6p^47s\ ^4p_{3/2}$	216.293	203-205	-3437(900)
$6p^5\ ^2p^o_{3/2} - 6p^47s\ ^4p_{3/2}$	216.293	207-205	2201(900)
$6p^5\ ^2p^o_{3/2} - 6p^47s\ ^4p_{3/2}$	216.293	209-205	6232(750)
$6p^5\ ^2p^o_{3/2} - 6p^47s\ ^4p_{3/2}$	216.293	211-205	9423(900)

Shift = ν (first isotope) − ν (second isotope).

²⁰⁵At hyperfine structure (*I* = 9/2)

Level	A (MHz)	B (MHz)	Shift of each <i>F</i> level from the centre of gravity (MHz)
$6p^5\ ^2p^o_{3/2}$	676(100)	1835(775)	F=3: -4735.96 F=4: -3255.29 F=5: -257.58 F=6: +5021.75
$6p^47s\ ^4p_{3/2}$	-365(3)	-367(50)	F=3: +2843.04 F=4: +1627.71 F=5: -120.83 F=6: -2555.50

Measured A, B constants; sources in the data file.